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BuddyBot: AI Powered Chatbot for Enhancing English Language Learning

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***Abstract*— Today, many students are having a hard time learning to speak the English language and the problem is even worse for those in rural areas, where both teachers and resources are rare. We are trying to break this disparity and provide better resources for those in need utilizing Artificial Intelligence and Natural Language Processing (NLP) to contribute to the SDG4 Quality and Inclusive Education. In this paper, we present the idea of how we developed Chatbot named BuddyBot Utilizing OpenAI Generative Pre-Trained Transformer 2 (GPT-2) and Google Text-to-Text Transfer Transformer (FLAN-T5) models. This system incorporates diverse datasets for fine-tuning and specialized tasks in English language learning. BuddyBot follows a five-phase architecture - User interface, Pre-processing, Natural Language Processing (NLP) stack, language generation engine and adaptive learning personalization. The E-Learn web-based interface offers interactive courses with BuddyBot assistance, emphasizing adaptive learning personalization. Dataset preprocessing includes tokenization, sentiment analysis, and specialized formatting. The NLP stack integrates GPT-2 and FLAN-T5 models into the framework, maintaining dialogue structure and personalization. Validation tests reveal model effectiveness, with reduced training loss (0.778 to 0.414 by epoch 8) and increased accuracy (0.810 training, 0.792 validation). Overall, BuddyBot's robust methodology and validation results showcase its efficacy in providing personalized English language learning experiences.**

Keywords*— *English Language Learning, Artificial Intelligence Chatbot, OpenAI GPT-2, Google Flan-T5, Natural Language Processing (NLP), SDG4.

I. INTRODUCTION

The rural students are dropping secondary or higher education because of a lack of quality teachers [1] to train them in the English language. Web based digital learning along with individualized learning [2] can support better teaching pedagogy [3]. Chatbots were developed to automate several tasks. For example, admission enquiry process chatbots are implemented using feedback enabled learning system [4], chatbot enabled smart tourism system [5]. Self-defending AI chatbots are used for anti-phishing to support innocent email users [6]. Conversational AI Chatbots are used for empowering self-help women group [7]. This work had developed a prototype website named E-learn along with Artificial Intelligence (AI) powered chatbot named BuddyBot to assist individuals in learning English. Covid-19 pandemic had forced rural students to drop out and work for jobs, providing e-learning platform can provide opportunities to upskill them [8]. This work helps in contributing to the Sustainable Development Goal 4 (SDG4), which is a commitment to inclusive and equitable quality education and promote life-long learning opportunities for all [9]. E-learn goes beyond mere accessibility, it offers a wealth of comprehensive learning materials ensuring every student has the tools and support they need to succeed in learning English language needed for improving their further education or career growth.

The heart of E-Learn lies in BuddyBot, a sophisticated Artificial Intelligence (AI) chatbot powered by fine-tuned Natural Language Processing (NLP) models. Unlike simple chatbots, BuddyBot transcends rudimentary conversation to deliver interactive exercises, real-time feedback on grammar and vocabulary, and personalized experiences in writing English sentences. By providing virtual assistance and personalized feedback, BuddyBot empowers students and promotes engagement, as mentioned in [10].

In the research, two powerful models, OpenAI's GPT-2 and Google's FLAN-T5, were utilized for advanced natural language processing tasks. Generative Pre-trained Transformer 2, GPT-2 is a language generation model developed by OpenAI [11]. It excels in open-ended text generation and can understand context in conversations. FLAN-T5 is a variant of the T5 (Text-To-Text Transfer Transformer) model from Google's FLAX library [12]. It excels in summarization and question-answering tasks. GPT-2 and FLAN-T5 are fine-tuned in our work on task-specific datasets like the 3k Conversations Dataset [13] and 16k Messenger-Like Conversations [14], tailoring the models to excel in generating chatbot responses and understanding user emotions. Evaluation involves validation metrics like training loss, accuracy, and rouge score [15], showcasing the models' effectiveness and improvement during training. Section2 describes Related works. Section3 illustrates the data and methodology. Section4 showcases the experimental prototype and results. Section5 concludes with possible future works. This work embodies a global vision, breaking down socioeconomic barriers to create an accessible English language learning experience.

II. RELATED WORKS

This section reviews existing research on AI-powered language learning platforms and chatbots, highlighting the limitations of these approaches and demonstrating how E-Learn addresses them and offers unique benefits.

A. Existing AI powered Language Learning Platforms

Several research studies have explored the use of AI in language learning platforms and chatbots. Some notable examples include:

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- **EnglishBot**: A mobile app incorporating a conversational agent with adaptive feedback for users to practice dialogues [16].
- **E-tutor**: An Intelligent Tutor Systems (ITS) for adult learners of German as a foreign language, providing metalinguistic feedback tailored to learner errors [17].
- **Nihongo-CALI and BANZAI:** An ITSs for adult Japanese learners providing intelligent metalinguistic feedback [18].
- **Cleverbot**: A chatbot engaging users in open-ended conversations and adapting to their conversational style [19].
- **Duolingo**: Duolingo is a gamified mobile app focusing on vocabulary and grammar drills [20].

While these platforms and chatbots demonstrate the potential of AI in language learning, they fall short in addressing the limitations mentioned in Table I.

TABLE I. COMPARISON OF E-LEARN WITH EXISTING AI PLATFORMS

Feature	Existing Platforms	E-learn
Language model	Limited language models	Fine-tuned GPT-2 and FLAN T5
Conversational Style	Static and Repetitive	Natural and Engaging
Feedback	Limited and static	Adaptive
Learning focus	Specific grammar rules or vocabulary	Conversational Fluency
Learning path	Fixed and determined	Adaptive and individualized
Engagement	Short-term and novelty-driven	High and sustained

E-Learn leverages the capabilities of two powerful models, GPT-2 and Flan T5, to enhance its functionality and provide users with a diverse range of features. It gives users freedom in their learning journey, avoiding the imposition of a fixed path and fostering a learning experience that is adaptable and personalized to individual preferences.

III. DATA AND METHODOLOGY

This section describes System Architecture and the Dataset used.

A. System Architecture

BuddyBot system architecture is shown in Fig. 1 with five phases starting from the user interface to adaptive learning personalization.

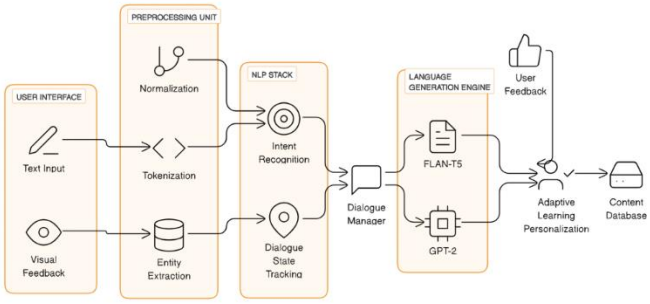


Fig. 1: System Architecture

These five phases are explained in brief.

1) **User interface:** It is a text-based interface where the user interacts with the BuddyBot
2) **Preprocessing Unit:** Implementing tokenization, The NLP object tokenizes the input text into individual tokens (words, sentences), Normalization: It lemmatizes each token (reducing words to their base or root form) and removes punctuation. Entity Extraction: It identifies named entities in the input. Named entities are phrases that refer to real-world objects.
3) **NLP Stack:** Intent recognition determines the user's intention or specific action for a given input. It involves identifying the request that the user wants to perform. Dialogue state tracking focuses on maintaining and updating the state of the ongoing conversation. It helps the system understand the user's queries in the context of the ongoing conversation and provides relevant responses. Dialogue Manager Keep track of the conversation history and current context to maintain a natural flow and avoid repetition and based on the Natural Language Understanding, NLU output, the dialogue manager determines the appropriate response or action to take.
4) **Language generation engine:** Employs fine-tuned GPT-2 and Flax T5 models for content creation. GPT-2 is trained on large web text corpora and generates engaging responses for chatbot interaction, while FLAN-T5 is trained on diverse text-to-text datasets encompassing various tasks and excels in summarization, and question answering.
5) **Adaptive Learning Personalization:** Buddy Bot learns from its interactions with users and improves its performance over time. Build user profiles based on their preferences, interactions, and past conversations to personalize future interactions. Regularly monitor the system's performance and user feedback to identify areas for improvement and update the models and algorithms accordingly.

B. Working Model of the System

The working model of the system is shown in Fig. 2. The model utilizes diverse datasets for raw data, pre-process and tokenize them. After gathering input sentences from the user, the following steps will be carried out in sequence:

- **Data Processing:** Preprocesses and tokenizes raw data, splitting it into Training and Testing sets for language model training.
- **Query Labeling:** Assigns labels (m1 or m2) to user queries, representing GPT-2 or FLAN-T5 model queries.
- **Model Training:** Trains GPT-2 and FLAN-T5 models separately using diverse datasets.
- **Model Selection:** Evaluates and selects the best-performing model based on predefined metrics.

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- **Model inference:** Processes tokenized input using the selected model (GPT-2 or FLAN-T5) to generate output.
- **Output Presentation:** Post-process model output for enhanced formatting and readability, presenting it through a user interface.

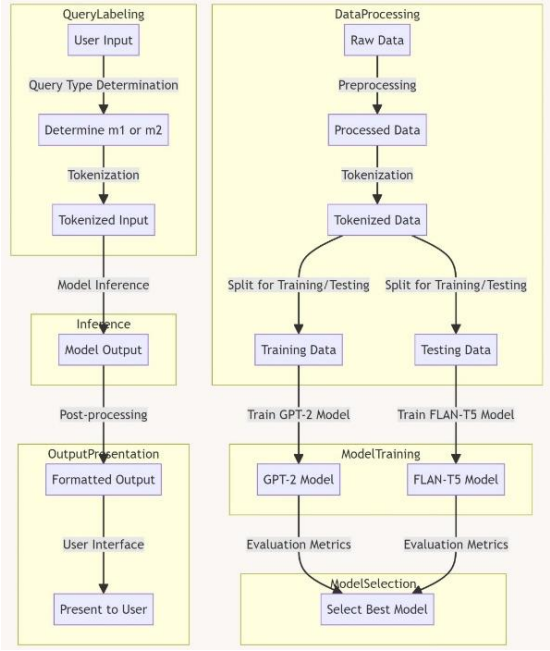


Fig. 2: System Workflow

C. Dataset

Dataset used are highlighted in Table II followed by brief technical explanations of it along with the preprocessing done in this work.

TABLE II. DATASET COMPOSITION

Dataset	Description	Purpose
3k-conversations-dataset-for-chatbot [13]	Contains dialogues used for fine-tuning models for generating chatbot responses	Fine-tuning the GPT-2 model for generating chatbot responses.
16k messenger-like conversations [14]	16,000 dialogues resembling real-world messenger exchanges	Training chatbots to understand the emotions; and opinions of users.
Tenses [21]	Transcript of tenses video tutorial with 3500 tenses sentences	Train the chatbot to generate sentences
big.txt [22]	Contains 1,28,457 English conversation dataset.	Train model to generate relevant sentences
word_meaning_ example dataset [23]	Contains 13,000+ words, it meanings and example	Train the model to generate vocabulary

- **3k Conversations dataset** [13]contains dialogues for fine-tuning GPT-2 and FLAN T5 model by Tokenization of dialogues and handling special characters and formatting for GPT-2 AND FLAN T5 input. 16k Messenger-Like Conversations:
- **16k Messenger-Like Conversations dataset** [14], 16,000 dialogues resembling real-world messenger exchanges used training chatbots to understand emotions and opinions of users by Sentiment analysis for emotion understanding. removing special characters, URLs, and timestamps.
- **Tenses Dataset** [21]: Transcript of tenses video tutorial with 3500 tenses sentences to train the chatbot to generate sentences by converting dataset into a question and answer format and handling spaces and punctuation.
- **big.txt dataset** [23], contains 1,28,457 English conversation dataset to train model to generate relevant sentences by removing punctuation, hyphens, and other unnecessary characters and Tokenization for sentence generation.
- **Word Meaning Example Dataset** [23], contains 13,000 words, their meanings, and examples to train the model to generate vocabulary by Tokenization of words and examples and removing spaces ensuring consistent formatting of meanings and examples.

IV. EXPERIMENTAL PROTOTYPE AND RESULTS

Project E-Learn integrates AI-powered chatbot technology into the E-Learn website for personalized English language learning. The system consists of BuddyBot AI chatbot built in E-Learn webpage.

A. E-learn interface: BuddyBot UI design

E-Learn offers users an interactive interface with BuddyBot conversations, learning materials, and progress tracking. Detailed dashboards and reports aid in monitoring individual learning journeys. Administrators benefit from a dedicated system management interface, allowing them to monitor performance, configure settings, and enhance the platform's library. The inclusion of user feedback provides valuable insights into user behavior, and preferences enabling administrators to personalize and optimize the platform.



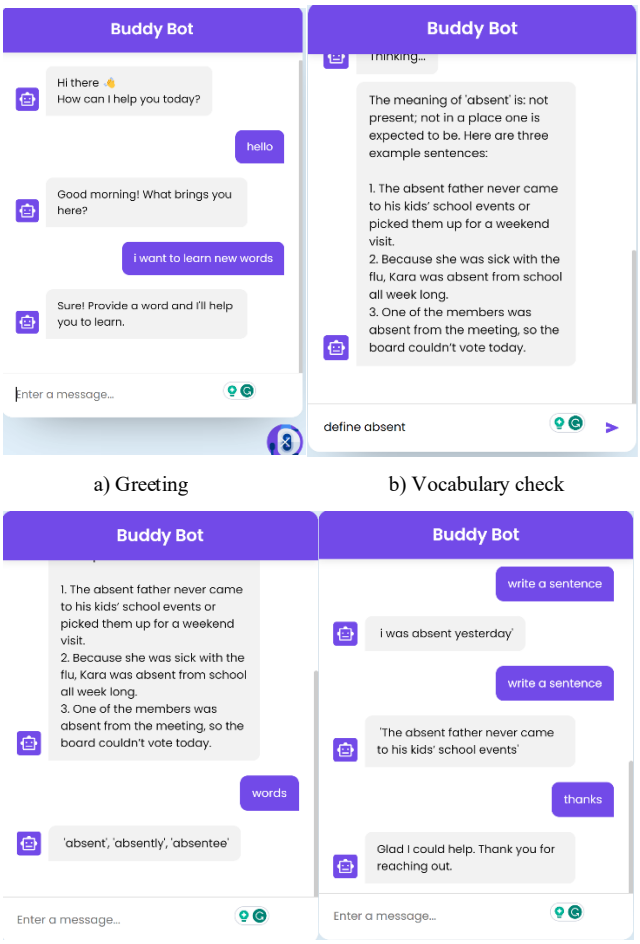
Fig. 2: E-Learn Interface

Fig. 2. shows the E-learn website, which offers a variety of courses to help people improve their English skills in essential areas, including grammar, speaking, writing, and reading. E-learn offers a chatbot called BuddyBot. Users can chat with BuddyBot about their English learning goals, ask questions, and get help with their writing, correcting the spelling and grammar in the sentences given by users.

BuddyBot employs FLAN-T5 for intelligent conversation, analyzing user intent, and tracking dialogue states. Real-time response generation utilizes a fine-tuned GPT-2 model for context-aware replies. The system offers grammar feedback for precise error correction. With

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personalized learning, Buddy Bot ensures goal-oriented dialogue, guiding users toward specific learning objectives like vocabulary building, grammar practice, or conversational fluency.



a) Greeting b) Vocabulary check

c) Synonyms d) Sentence generation

Fig. 3: BuddyBot User Interface (UI)

Fig. 3 depicts the User Interface (UI) and working of the BuddyBot with an example conversation between a user and buddy bot. Fig. 3a) is initiating the conversation. Fig. 3b) shows the user asking Buddybot for 'absent' meaning and Buddybot gives the word meaning with examples. Fig. 3c) shows the user asks for similar words then buddy bot gives possible synonym of words. Fig. 3d) shows the user asks for a sentence and Buddybot generates the relevant sentence using the given word.

B. Dialogue Manager Implementation

Dialogflow framework [24] is used, to define dialogue structure, including intent recognition, and identify the relevant actions, and map the potential conversation paths and user responses. Model trains on diverse datasets 3k Conversations [13], 16k Messenger-Like Conversations [14](sentiment analysis), Tenses Dataset [21] (Q&A format), big.txt (sentence generation) [22], and Word Meaning Example Dataset (vocabulary)[23] as mentioned in Table II.

In the GPT-2 fine-tuning process, hyperparameters, including 4 epochs, batch size 4/GPU, learning rate 5e-5, and 8 gradient accumulation steps, are meticulously configured with TrainingArguments for memory efficiency. Tuned save_steps, evaluation_strategy, and eval_steps regulate checkpoint frequency and validation intervals, enhancing model precision during training.

FLAN-T5 fine-tuning utilizes the 'google/flan-t5-base' model and tokenizer. The tokenize_function converts conversation pairs to LM instructions, managed by Trainer class with specified parameters. PEFT adapts FLAN-T5 through Low-Rank Adaptation (LoRA) and adapter layers, reducing parameters for efficiency. FLAN-T5 focuses on Parameter Efficient Fine-Tuning(PEFT) with LoRA [25], while GPT-2 employs conventional strategies in parameter tuning.

The system then integrates GPT-2 and FLAN-T5 deep learning models into the framework. GPT-2 excels in open-ended text generation [11], while FLAN-T5 is more versatile [12], handling diverse tasks effectively. Flan T5 model achieved the best rouge score of 47.57 and T5 model achieved a rouge score of 23.82 on the test set is illustrated in Fig. 4. Thus, Flan T5 model is performing better than T5 base model and thus this model is chosen for the system.

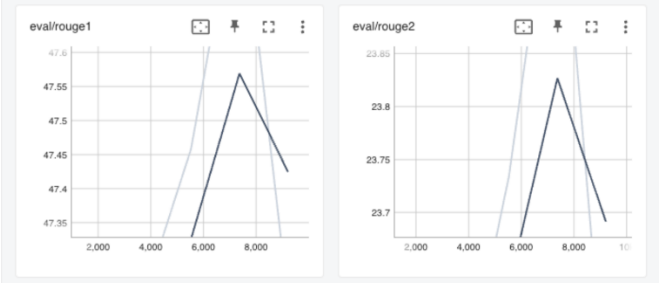


Fig. 4. Rogue score a) FLAN T5 b) T5

The spell correction module implemented using the Spello library [22], operates by training on large text data consisting of 128457 sentences. The training involves three step processes, including context model, Symspell, and Phoneme training. The model learns patterns and relationships between words to identify and correct spelling errors. During spell correction, the model analyzes input sentences, identifies misspelled words, and suggests appropriate corrections. Sample Spell correction results is mentioned in Fig. 5.

```
DEBUG:spello.settings:Spell-correction Results {'origina
{'original_text': 'I want to paly mony',
 'spell_corrected_text': 'I want to pay money',
 'correction_dict': {'paly': 'pay', 'mony': 'money'}}
```

Fig. 5: Spell correction results.

C. Validation Test results

```
all_acc['train_acc'].append(train_acc)
all_acc['val_acc'].append(val_acc)

print(f'Epoch: {epoch}, train_loss: {torch.tensor(train_loss).mean():.3f}, train_a

Epoch: 0, train_loss: 0.778, train_acc: 0.557, val_loss: 0.641, val_acc: 0.607
Epoch: 1, train_loss: 0.591, train_acc: 0.692, val_loss: 0.519, val_acc: 0.764
Epoch: 2, train_loss: 0.516, train_acc: 0.757, val_loss: 0.526, val_acc: 0.766
Epoch: 3, train_loss: 0.475, train_acc: 0.788, val_loss: 0.484, val_acc: 0.787
Epoch: 4, train_loss: 0.452, train_acc: 0.794, val_loss: 0.475, val_acc: 0.787
Epoch: 5, train_loss: 0.440, train_acc: 0.798, val_loss: 0.498, val_acc: 0.787
Epoch: 6, train_loss: 0.432, train_acc: 0.805, val_loss: 0.476, val_acc: 0.794
Epoch: 7, train_loss: 0.425, train_acc: 0.810, val_loss: 0.476, val_acc: 0.793
Epoch: 8, train_loss: 0.414, train_acc: 0.813, val_loss: 0.492, val_acc: 0.791
Epoch: 9, train_loss: 0.420, train_acc: 0.810, val_loss: 0.489, val_acc: 0.792
```

Fig. 6. Validation results

The model demonstrates improvement during training, evident by a decrease in loss and increased accuracy. Training loss starts at 0.778, dropping to 0.414 by epoch 8 is presented

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in Fig. 6. Both training and validation accuracy steadily increase, reaching 0.810 and 0.792, respectively, indicating effective learning and generalization without overfitting.

Fig. 7 shows multi-plot visualization to showcase the training and validation metrics (loss and accuracy) of the model during the training process. The first plot in Fig. 7 a) displays the training loss values over different epochs. It gives insights into how well the model is minimizing its error on the training dataset. The second Plot in Fig. 7 b) illustrates the validation loss values over epochs. It provides information on how well the model generalizes to unseen data during the training process. The third plot in Fig. 7c) represents Training and Validation Accuracy. The x-axis label is set to 'epoch', and the y-axis label is set to 'accuracy'. It consists of two lines representing training and validation accuracy over epochs. This plot helps to assess how well the model is learning to make correct predictions on both the training and validation datasets.

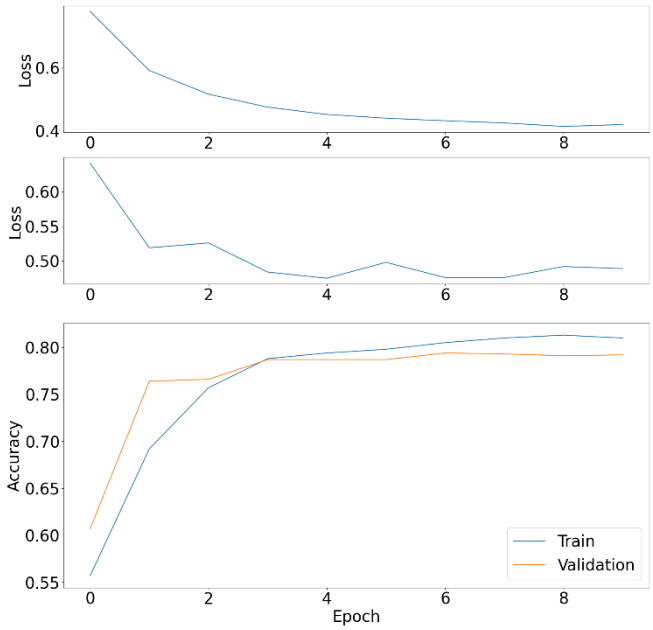


Fig. 7: Performance Evaluation a) Training Loss, b) Validation Loss, c) Training and Validation Accuracy

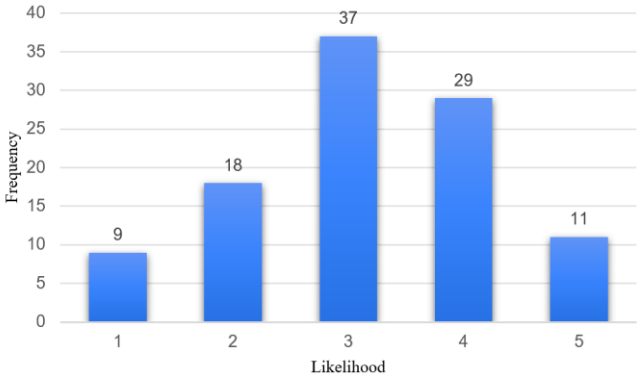
Table III Language Benchmark

Task	Accuracy	Benchmark
Short Dialogues (2-5 turns)	85%	ROUGE-1: 42.5, ROUGE-2: 25.6
Longer Dialogues (6-10 turns)	78%	ROUGE-1: 38.2, ROUGE-2: 22.4
Factual Question Answering	83%	F1-score: 78.4
Open-ended Question Answering	76%	BLEU score: 62.4
Dialogue Generation	80%	Human Evaluation: 3.8/5 fluency, 3.6/5 coherence
Semantic Analysis	82%	Accuracy: 85% (sentiment), 79% (intent)

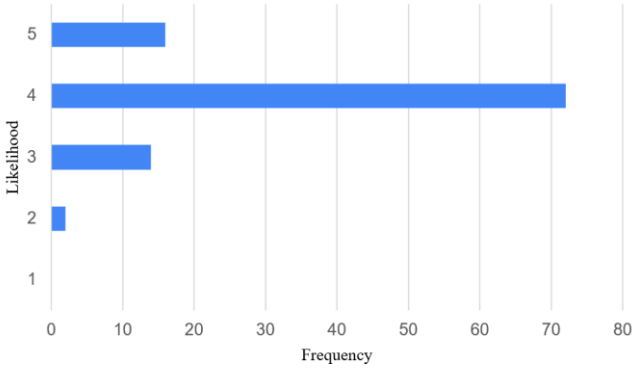
The model excels across diverse English Language learning tasks: achieving 80% accuracy in dialogue generation, demonstrating proficiency in question answering with an 83% accuracy, and obtaining a Bi-lingual Evaluation Under study (BLEU) value[26] with an accuracy of 72.4%. The other language benchmarks are provided in Table III.

D. Quantitative survey

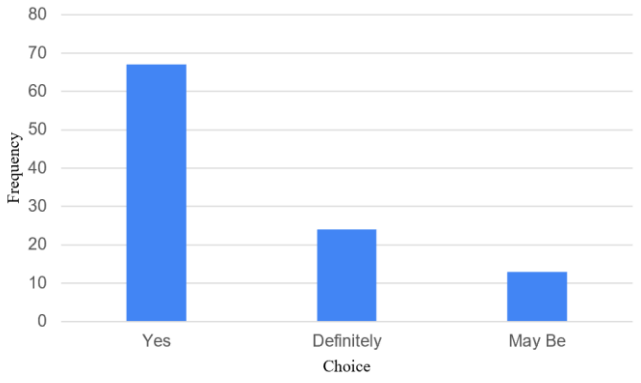
To assess the effectiveness of AI chatbot, a quantitative survey was conducted using google forms involving a total of 104 participants after they had used the built BuddyBot chatbot.



a) AI chatbot usage



b) Likelihood of future usage



c) Chatbot recommendation

Fig. 8: Quantitative survey results of 104 participants

The survey aimed to gather insights into the users' experiences with the chatbot, their perceptions of its effectiveness, and their likelihood of continuing its usage in the future. Among the 104 participants, 27 participants (8.7+17.3=26%) reported not regularly using the chatbot mentioned for survey question “How frequently do you use the AI chatbot for English Learning?” in Fig. 8a. Despite their infrequent usage of AI chatbot, they found the chatbot like BuddyBot can be effective when they did use it.

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Approximately 69.2% indicated being very likely to continue using the chatbot in the future for the survey question "How likely are you to continue using the AI chatbot for English learning in the future?" is shown in Fig. 8b. 87.5% expressed a willingness to recommend the chatbot for English learning to others is mentioned in Fig. 8c.

These findings indicate a favorable user perception and highlight the potential for AI-powered chatbots in role in language learning, providing personalized feedback and an enjoyable learning environment for users.

V. CONCLUSION AND FUTURE WORKS

In conclusion, this research highlights the impact of E-Learn and BuddyBot in revolutionizing AI in English language education. By harnessing the capabilities of Open AI GPT-2 and FLAN T5, the study exemplifies the potential of personalized and interactive learning. E-Learn not only overcomes digital barriers but also redefines online education, offering a comprehensive and accessible platform.

Moving forward, the model can focus on enhancing accuracy, contextual understanding, and speech recognition. User feedback integration and continuous alignment with NLP advancements will drive model improvement. To enhance inclusivity, future iterations of E-Learn will incorporate audio input for regional languages expanding accessibility for diverse learners.

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